

3.9 Organic Synthesis

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.9 Organic Synthesis
Difficulty	Hard

Time allowed: 40

Score: /26

Percentage: /100

Question 1a

This question is about a reaction pathway to produce 1-bromo-1-methylcyclohexane.

1-bromo-1-methylcyclohexane can be produced by the following reaction steps, as shown in Fig. 1.1 below.



Fig. 1.1

Name and outline the mechanism for the reaction between compound **E** and hydrogen bromide, HBr. In your answer, draw the major product of the reaction.

[6 marks]

Question 1b

Explain why a major product and minor product are produced in the reaction outlined in part (a)

[3 marks]

Question 1c

Suggest a possible starting molecule for the reaction pathway in part (a) and give the reagents required for step 1 based on this starting molecule.

[2 marks]

Question 2a

This question is about reaction profiles using chloroalkanes

For the reaction profile outlined in Fig. 2.1, state the mechanism or type of reaction for each step.



Fig. 2.1

[2 marks]

Question 2b

Step 2 is carried out by heating under reflux.

Identify Compound H.

[1 mark]

Question 2c

1-chlorobutane can be produced from an alkene. Identify the alkene and give the reagent required.

[2 marks]

Question 2d

Explain how the functional group of the final compound in the reaction scheme in part (a) would be different if 2-chlorobutane was used as the starting compound.

[2 marks]

Question 3a

Compounds **W**, **X** and **Y** are all compounds containing carbon, hydrogen and oxygen only. Compounds **X** and **Y** are straight chain compounds, each containing five carbons. Compound **W** and a byproduct, compound **Z**, are formed from the reaction between compound **X** and **Y**. Compound **X** can be oxidised by the reaction with acidified potassium dichromate to give compound **Y**.

2.754 g of compound **Y** contains 0.027 moles.

Using the information given, state the name of compound **Y**.

[2 marks]

Question 3b

Deduce the structural formula of compound **W** and explain how compound **Z** is formed in the reaction.

[2 marks]

Question 3c

Compound **X** will oxidise to compound **Y** if allowed to fully oxidise. Explain how a student could stop the full oxidation of compound **X**.

[2 marks]

Question 3d

Deduce the formula of an isomer of compound **X** that will not react with acidified potassium dichromate, $\text{H}^+ / \text{K}_2\text{Cr}_2\text{O}_7$.

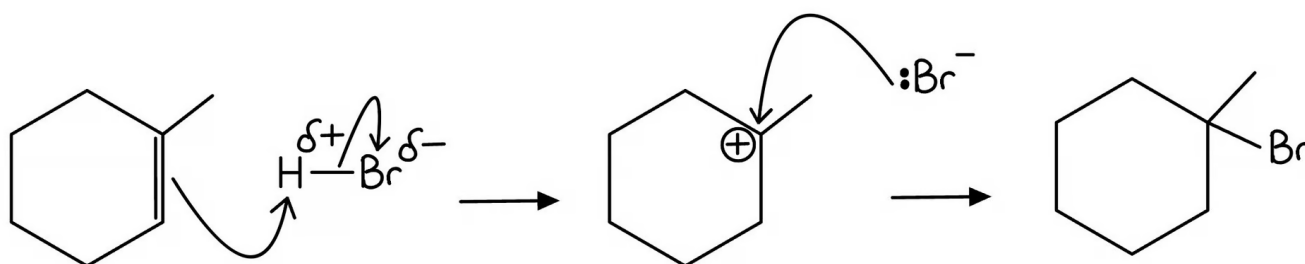
[2 marks]

Model Answers: Hard

1a

a) The name and mechanism for the reaction between compound **E** and hydrogen bromide, HBr is:

- Electrophilic addition; [1 mark]



- Curly arrow from C=C to H in H-Br
AND
 δ^+ H and δ^- Br shown; [1 mark]
- Curly arrow from H-Br bond to Br atom; [1 mark]
- Tertiary carbocation identified and labeled; [1 mark]
- Curly arrow from lone pair in $:\text{Br}^-$ to carbocation; [1 mark]
- Correct structure of the major product; [1 mark]

[Total: 6 marks]

- The methyl group next to the C=C double bond will mean a tertiary carbocation can be produced
 - These are more stable than secondary carbocations
 - Therefore the major product will be 1-bromo-1-methylcyclohexane

1b

b) A major and minor product are produced because:

- The major product is formed as the intermediate would contain a tertiary carbocation; [1 mark]
- The minor product is produced as the intermediate would contain a secondary carbocation; [1 mark]
- Tertiary carbocations are more likely to bond with the Br atom as they are more stable

OR

Secondary carbocations are more likely to bond with the H atom as they are less stable; [1 mark]

[Total: 3 marks]

- This is due to Markovnikov's rule regarding addition across the C=C of an asymmetric alkene
 - The H-Br will add across the double bond (C=C)
 - The H atom will most likely form a bond with the least substituted carbon atom leaving a tertiary carbocation
 - This tertiary carbocation will form a bond with the Br atom

1c

c) A possible starting molecule is:

- 2-methylcyclohexanol

OR

1-methylcyclohexanol; [1 mark]

The reagent required is:

- Al_2O_3

OR

Conc. acid; [1 mark]

[Total: 2 marks]

- The question asks to suggest a possible starting molecule, however, based on the reactions you are expected to know at this level, it must be an alcohol
- The only reaction you know which produces an alkene is the dehydration / elimination of an alcohol

- The alcohol group can be placed on either side of where the double bond should be
- You are not expected to know the reaction mechanism for this type of reaction

2a

a) The mechanisms are:

- Step 1 = nucleophilic substitution; [1 mark]
- Step 2 = oxidation; [1 mark]

[Total: 2 marks]

- You should recognise that the reagents $K_2Cr_2O_7 / H_2SO_4$ in step 2 are used to oxidise an alcohol
- This then allows you to figure out that step 1 is going from a halogenoalkane to an alcohol, which is nucleophilic substitution

2b

b) Compound H is:

- Butanoic acid; [1 mark]

[Total: 1 mark]

- As the starting compound is 1-chlorobutane then the alcohol formed will be a primary alcohol, butan-1-ol
- As step 2 is carried out by heating under reflux then the resulting compound will be a carboxylic acid
- If this step was carried out using distillation an aldehyde would have been produced

2c

c) The alkene required is:

- But-1-ene; [1 mark]

The reagent required is

- Hydrogen chloride; [1 mark]

[Total: 2 marks]

- 1-chlorobutane is the minor product made when but-1-ene, an unsymmetrical alkene, undergoes electrophilic addition
 - 2-chlorobutane would be the major product formed
- But-2-ene and HCl will only produce 2-chlorobutane as a product

2d

d) The functional group of the final compound in the reaction scheme in part (a) would be:

- A ketone (rather than a carboxylic acid); [1 mark]
- (As) secondary alcohols are oxidised to ketones when heating under reflux; [1 mark]

[Total: 2 marks]

- Using the reaction scheme given in part (a), if 2-chlorobutane is used then reaction step 1 will result in a secondary alcohol - butan-2-ol
- Heating a secondary alcohol under reflux will result in a ketone being produced

3a

a) To work out the identity of compound Y:

- $M_r(\text{compound Y}) = \frac{2.754 \text{ g}}{0.027 \text{ moles}} = 102.00 \text{ g mol}^{-1}$; [1 mark]
- Pentanoic acid; [1 mark]

[Total: 2 marks]

- There is a lot of information given in the question and the first step will be to calculate the molar mass of compound Y
- You know that it contains 5 C atoms so that is $5 \times 12 = 60 \text{ g mol}^{-1}$ of the molar mass accounted for, leaving 42 g mol^{-1}
- You are aware that it is an oxidation product so must contain at least 1 O atom

- $M_r(\text{compound Y}) = \frac{2.754 \text{ g}}{0.027 \text{ moles}} = 102.00 \text{ g mol}^{-1}$; [1 mark]
- Pentanoic acid; [1 mark]

[Total: 2 marks]

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- You know that it contains 5 C atoms so that is $5 \times 12 = 60 \text{ g mol}^{-1}$ of the molar mass accounted for, leaving 42 g mol^{-1}
- You are aware that it is an oxidation product so must contain at least 1 O atom
 - If 1 O atom was present, 26 g mol^{-1} would be due to H, which is not feasible
 - If 2 O atoms are present, 10 g mol^{-1} would be due to H, which gives a credible product
- This gives the formula to be $\text{C}_5\text{H}_{10}\text{O}_2$
- Carboxylic acids contain 2 O atoms and are obtained through the oxidation of primary alcohols when refluxed with acidified potassium dichromate

3b

b) The formula for compound **W** and formation of compound **Z** is:

- Compound **W** is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

OR $\text{C}_{10}\text{H}_{20}\text{O}_2$; [1 mark]

- Compound **Z** is water / H_2O

AND

Produced by a condensation reaction between pentanoic acid and pentanol / compounds **X** and **Y**; [1 mark]

[Total: 2 marks]

- You should be aware that the reaction between an alcohol and carboxylic acid produces an ester and water
 - In this reaction, concentrated sulfuric acid, H_2SO_4 , is used as a catalyst
- The formation of the ester eliminates a water molecule through a condensation reaction
 - **Careful:** A condensation reaction does **not** mean that water is formed; it actually means that a small molecule is lost
- The name of compound **W** is pentyl pentanoate

3c

c) A student could prevent the full oxidation of compound **X** by:

- Distill off aldehyde as soon as it is produced; [1 mark]
- Perform the reaction above the boiling point of the aldehyde / pentanal; [1 mark]

[Total: 2 marks]

- The full oxidation of a primary alcohol such as Compound **X** will produce
 - An aldehyde followed by a carboxylic acid
- If we want to prevent this full oxidation, we **must** alter the reaction conditions and use different apparatus
 - A Liebig condenser is used for this purpose
 - This piece of apparatus is ideal for the separation of an organic product from its reacting mixture
 - It allows the distillate to be collected in an approximate boiling point range

3d

d) The formula of the isomer and the reason it can not react with acidified potassium dichromate, $\text{H}^+ / \text{K}_2\text{Cr}_2\text{O}_7$ is:

- $\text{CH}_3\text{C}(\text{CH}_3)\text{OHCH}_2\text{CH}_3$; [1 mark]
- It doesn't react because it is a tertiary alcohol; [1 mark]

[Total: 2 marks]

- Earlier, compound **X** was identified to be an alcohol with 5 carbons, which gives it the molecular formula $\text{C}_5\text{H}_{11}\text{OH}$
- For the isomer to not react with acidified potassium dichromate, it must be a tertiary alcohol
- If you are unsure it is best to draw out the molecule
- The only isomer which gives a tertiary alcohol is 2-methylbutan-2-ol